

SHOULD-COST ANALYSIS REPORT

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Bottom-Up Manufacturing Cost Model · Fully Traceable Rate Data

Seat Locking Bracket

Commodity: SHEET METAL · Currency: CNY · FX Rate: 9.0498 to GBP · Operations: 1 · Region: CN

Total Should-Cost	Material	Process	Labour	Margin
¥15.87	24.1%	0.7%	0.1%	7.4%

Model Confidence: **Low** · 1 traced operations · 4 data points auditable

Engine Warnings (1)

rawMaterial.materialId: Material rate confidence: Medium

Geometry Provenance — MEASURED (OCCT B-rep kernel)

Volume, weight and every feature were measured from the CAD solid by the Open CASCADE kernel. Measured volume 71.1 cm³; mass for the selected material family 0.558 kg.

Material cost and geometry-derived machining are grounded in the actual solid — not an estimate.

Key Assumptions

Annual volume: 200,000 · Region: CN · Commodity: sheet metal

Alloy / material: DC01 · Net weight: 0.558 kg (measured)

Material utilisation: 100% · Overhead: 12% of factory base · Margin: 8% of subtotal · Operations: 1

§1 — 8-Bucket Cost Breakdown

Cost Bucket	Amount (CNY)	% of Total	Cost Mix Bar
1. Raw Material	¥3.82	24.1%	
2. Process (Machine)	¥0.11	0.7%	
3. Direct Labour	¥0.02	0.1%	
4. Tooling (amortised)	¥7.88	49.7%	
5. Packaging	¥0.63	4.0%	

Cost Bucket	Amount (CNY)	% of Total	Cost Mix Bar
6. Logistics	¥0.81	5.1%	
Factory Cost	¥13.27	83.6%	
7. Overhead (SG&A)	¥1.42	8.9%	
Subtotal	¥14.69	92.6%	
8. Supplier Margin	¥1.18	7.4%	
TOTAL SHOULD-COST	¥15.87	100.0%	

§2 — Commercial Parameters

Overhead Rate	12.0%	Supplier Margin	8.0%
Packaging / part	¥0.63	Logistics / part	¥0.81
Total Tooling Cost	¥1576325.79	Amortisation Volume	200,000 parts

§3 — Material Detail

Parameter	Value	Unit	Notes
Material ID	mat-dc01	ID	
Grade / Specification	DC01		UK CR mild sheet, delivered small-lot. Index-anchored: C R C € 7 8 1 / t ! ' £ 0 . 6 7 / k g m i l l + £ 0 . 1 9 / k g s t o c k h o u s e p r e m i u m . B a s i s : F a s t m a r k e t s E U H R C € 6 9 1 / t M a y - 2 6 + € 9 0 C R C , F X 0 . 8 5 5 . Y i e l d ~ 1 4 0 M P a . R e g i o n a l a d j . × 0 . 8 8 0 (m i l l S t e e l)
Region	China		
Net Finished Weight	0.5580 kg	kg	Weight in finished part
Gross Weight (stock)	0.5580 kg	kg	Net ÷ utilisation ratio
Scrap / Runner Weight	0.0000 kg	kg	Gross - Net
Material Utilisation	100.0%	%	Benchmark: stamping/blanking 55–75 % (nesting + skeleton scrap)
Material Price	¥6.85	CNY /kg	UK CR mild sheet, delivered small-lot. Index-anchored: C R C € 7 8 1 / t ! ' £ 0 . 6 7 / k g m i l l + £ 0 . 1 9 / k g s t o c k h o u s e p r e m i u m . B a s i s : F a s t m a r k e t s E U H R C € 6 9 1 / t M a y - 2 6 + € 9 0 C R C , F X 0 . 8 5 5 . Y i e l d ~ 1 4 0 M P a . R e g i o n a l a d j . × 0 . 8 8 0 (m i l l S t e e l)
Scrap Recovery Price	¥1.59	CNY /kg	
Gross Material Cost	¥3.82	CNY	Gross × price/kg
Scrap Credit	-¥0.00	CNY	Scrap × recovery price
NET RAW MATERIAL COST	¥3.82	CNY	Gross - scrap credit + consumables
Data Confidence	Medium		
Effective Date	2026-07		

§4 — Operations Detail

4A Machine Operations

Operation	Machine Class	Rate / hr	Cycle (min)	Parts / Cycle	OEE %	Machine Cost
Press (transfer)	100T Mechanical Press (general press shop)	¥72.91	0.07	1	85.0%	¥0.11
TOTAL						¥0.11

4B Labour Detail

Operation	Labour Grade	Rate / hr	Man ning	Lab. min	Eff %	Labour Cost	Op Total
Press (transfer)	Semi-skilled Operator	¥49.77	0.25	0.07	95.0 %	¥0.02	¥0.12
TOTAL						¥0.02	¥0.12

4C Machined Features — Geometry Audit

Feature	Ø×depth / area	Qty	Operation	Min	Cost	Costed ?
Face	13125 mm ²	2	Face milling (facing)	3.68	¥15.60	Yes
Face	7909 mm ²	2	Face milling (facing)	2.38	¥10.07	Yes
Face	7170 mm ²	2	Face milling (facing)	2.19	¥9.29	Yes
Face	2951 mm ²	2	Face milling (facing)	1.14	¥4.82	Yes
Face	591 mm ²	2	Face milling (facing)	0.55	¥2.32	Yes
Pocket	13125 mm ² × 15	2	Pocket milling (rough + finish)	66.03	—	No
Pocket	7909 mm ² × 18	1	Pocket milling (rough + finish)	24.43	—	No
Pocket	7909 mm ² × 12	1	Pocket milling (rough + finish)	15.80	—	No
Pocket	7170 mm ² × 22	1	Pocket milling (rough + finish)	27.09	—	No
Pocket	7170 mm ² × 8	1	Pocket milling (rough + finish)	9.47	—	No
Pocket	2951 mm ² × 22	1	Pocket milling (rough + finish)	11.67	—	No
Pocket	2951 mm ² × 7	1	Pocket milling (rough + finish)	3.92	—	No
Pocket	591 mm ² × 16	1	Pocket milling (rough + finish)	2.00	—	No
TOTAL costed		10		9.94	¥42.12	Yes

9 detected feature(s) NOT costed

9 machinable feature instance(s) (160.4 min) were detected in the geometry but are NOT included in this cost — on a near-net part they may be as-cast. If they are machined, tick them in the Machined-Features panel and re-cost; this could add up to ¥679.89.

\$5 — Machine Rate Buildup

100T Mechanical Press (general press shop) [press-100t] · Confidence: Low · UK general press-shop benchmark (standard/amortised mechanical press), Jun 2026 | Regional: capex/overhead ×0.55, energy re-tariffed @£0.07/kWh

Cost Component	Annual Cost (CNY)	Rate/hr @80% util	Notes
Depreciation	¥89592.76	¥32.00	3,500 hr/yr available
Maintenance	¥44796.38	¥16.00	
Energy	¥13771.39	¥4.92	
Floor Space	¥24886.88	¥8.89	
Indirect Support	¥19909.50	¥7.11	
Finance Cost	¥11199.10	¥4.00	
TOTAL — 100T Mechanical Press (general press shop)	¥204156.01	¥72.91	Effective hours: 2800/yr

How to read the machine rate

Rate/hr = (annual depreciation + maintenance + energy + floor + indirect + finance) ÷ effective hours, where effective hours = available hours × utilisation.
Effective hours encode the shift pattern: a lower hours-base raises £/hr and a higher one lowers it. A challenge from a supplier will target this divisor and whether depreciation is machine-only or full-cell (machine + dosing furnace + trim + robots/extraction) — state your basis when defending the number.

\$6 — Rate Traceability

Field	Value	Unit	Source / Reference	Rate ID	Conf.
material.pricePerKg	0.7568	£/kg	UK CR mild sheet, delivered small-lot. Index-anchored: CRC €781/t! £0.67/kg mill + £0.19/kg stockholder premium. Basis: Fastmarkets EU HRC €691/t May-26 +€90 CRC, FX 0.855. Yield ~140 MPa. Regional adj. ×0.880 (millSteel)	mat-dc01	Medium
material.scrapRecoveryPricePerKg	0.1760	£/kg	UK CR mild sheet, delivered small-lot. Index-anchored: CRC €781/t! £0.67/kg mill + £0.19/kg stockholder premium. Basis: Fastmarkets EU HRC €691/t May-26 +€90 CRC, FX 0.855. Yield ~140 MPa. Regional adj. ×0.880 (millSteel)	mat-dc01	Medium
Press (transfer).machineRatePerHr	8.0569	£/hr	UK general press-shop benchmark (standard/amortised mechanical press), Jun 2026 Regional: capex/overhead ×0.55, energy re-tariffed @£0.07/kWh	press-100t	Low
Press (transfer).labourRatePerHr	5.5000	£/hr	Regional benchmark China — 2026 Q2	lab-uk-semiskilled	Low

§7 — Should-Cost with Uncertainty

Monte-Carlo · Low confidence

Estimate (± band)	P10 · optimistic	P50 · median	P90 · conservative	Band	CV
¥15.87 ± 29.8%	¥11.49	¥15.20	¥21.00	wide	24.3%

Range reflects how well each cost driver is known (High / Medium / Low confidence). Tighten it by attaching CAD, a BOM, or logging an actual quote.

Why confidence is Low

Main band drivers: 2 rate(s) at Low confidence; tooling amortisation carries the widest per-bucket spread; some detected features not costed (see §4C).

What's excluded from this unit cost

One-time NRE — tooling / die, fixtures and CNC programming — is amortised separately, not in this unit cost.
 Import duty and international freight (regional table is Ex-Works).
 Formal embodied-carbon reporting (figures are indicative cradle-to-gate — replace with supplier EPDs).

§8 — Sensitivity Analysis

± 10% driver swing · ranked by ¥ impact

Cost Driver	Base	-10% cost	+10% cost	Range ¥
Amortisation Volume	200000parts	+6.7%	-5.5%	¥1.93
Tooling Cost	¥1576325.79	-6%	+6%	¥1.91
Material: DC01	¥6.85/kg	-2.9%	+2.9%	¥0.92
Material utilisation	1	+2.5%	0%	¥0.39
Overhead %	12%	-1%	+1%	¥0.31
Supplier Margin %	8%	-0.7%	+0.7%	¥0.24
Press (transfer): Cycle Time	0hr	-0.1%	+0.1%	¥0.03
Press (transfer): Machine Rate	¥72.91/hr	-0.1%	+0.1%	¥0.03

§9 — Regional Cost Comparison

Ex-Works · per-region should-cost · vs China

Region	Material	Process	Labour	Tooling	Overhead	Ex-Works	Logistics	Total	vs China
United Kingdom (GBP)	¥4.34	¥0.19	¥0.06	¥14.33	¥1.89	¥20.82	¥0.56	¥23.46	+48%
Germany (EUR)	¥4.47	¥0.20	¥0.10	¥15.05	¥2.18	¥22.00	¥0.65	¥24.84	+57%
France (EUR)	¥4.43	¥0.18	¥0.07	¥13.18	¥1.88	¥19.74	¥0.65	¥22.46	+42%
Spain (EUR)	¥4.34	¥0.17	¥0.04	¥12.61	¥1.63	¥18.80	¥0.65	¥21.44	+35%
Poland (PLN)	¥4.21	¥0.14	¥0.03	¥10.32	¥1.25	¥15.95	¥0.67	¥18.35	+16%
Turkey (TRY)	¥3.91	¥0.12	¥0.01	¥8.60	¥0.99	¥13.62	¥0.73	¥15.87	+0%
China (CNY)	¥3.82	¥0.11	¥0.02	¥7.88	¥1.42	¥13.25	¥0.81	¥15.87	Base
India (INR)	¥3.91	¥0.10	¥0.01	¥7.45	¥0.83	¥12.30	¥0.84	¥14.45	-9%
Mexico (MXN)	¥4.13	¥0.12	¥0.02	¥8.60	¥1.00	¥13.86	¥0.76	¥16.10	+1%
United States (USD)	¥4.34	¥0.17	¥0.08	¥12.18	¥1.59	¥18.36	¥0.70	¥20.97	+32%

Per-region should-cost using regional labour, energy and rate benchmarks. Tooling held fixed. Ex-Works — excludes import duty + international freight. Indicative — confirm with RFQ.

§10 — Embodied Carbon

1.37 kgCO2e · 2.45 /kg · cradle-to-gate

Material	1.17 kgCO2e	Material factor	2.10 kg/kg (Steel)
Process	0.11 kgCO2e	Grid intensity	0.550 kg/kWh · 0.20 kWh
Logistics	0.09 kgCO2e	Total	1.37 kgCO2e

Indicative cradle-to-gate factors — replace with supplier EPDs for formal reporting.

[WARNING] Tooling amortisation is a significant cost driver

Finding	Tooling at 49.7% (£0.87/part) exceeds the benchmark of 10%. This is volume-sensitive.
Impact	High · up to 10% potential saving
Action 1	Doubling volume to 400,000 parts halves tooling cost to £0.44/part
Action 2	Evaluate family tooling (multiple parts per die) to spread NRE across higher volumes

[OPPORTUNITY] Packaging & logistics cost is above average

Finding	Packaging + logistics at 9.1% of total cost. For high-volume parts this should be under 2%.
Impact	Low · up to 2% potential saving
Action 1	Evaluate returnable/reusable packaging (racks, dunnage trays) vs expendable packaging
Action 2	Consolidate shipment frequency — larger drops reduce per-unit logistics cost

[INFO] Material cost is below benchmark range

Finding	Material at 24.1% is below the typical 30–52% range. Conversion cost is the dominant driver.
Impact	Low
Benchmark	Material %: yours 24.1% vs industry 30–52%
Action 1	Focus optimisation efforts on process and labour efficiency

§12 — Cost-Reduction Opportunities (ranked by saving)

6 opportunities · combined ¥1.89/part (~12%)

Ranked on money per part against this costing. The combined figure is the root-sum-square of the top three, capped at 40% — overlapping actions do not save twice, so the column is deliberately not summed.

#	Category	Action	Why it is on the list	Save/p art	%	Timeframe	Risk
1	Tooling	Volume Increase to Dilute Tooling NRE	Tooling at 49.7% of part cost is volume-sensitive. Doubling volume halves per-part tooling cost.	¥1.59	10.0 %	Quick Win	Low
2	Tooling	Tooling Ownership and End-of-Life Terms	£174,184 of tooling sits behind this part. Who owns it, who maintains it, and what happens at end of programme are worth points of piece price — and all of it is negotiable before the tool is cut.	¥0.63	4.0%	Quick Win	Low
3	Tooling	Tool-Life Extension Programme	Tooling at 49.7% of part cost means every extra thousand shots of die life is money. Surface treatment and disciplined	¥0.63	4.0%	Medium Term	Low

#	Category	Action	Why it is on the list	Save/p art	%	Timeframe	Risk
			maintenance are the cheapest capacity there is.				
4	Design & Geometry	Design Review for DFM Compliance	Formal DFM review with manufacturing engineering to identify part features that increase cost without functional benefit.	¥0.79	5.0%	Quick Win	Low
5	Commercial & Sourcing	Annual Volume Re-Commitment for Better Pricing	Provide 12-month rolling volume forecast to supplier to secure better unit pricing and tooling amortisation.	¥0.48	3.0%	Quick Win	Low
6	Packaging & Logistics	Pack Density / Cube Utilisation Review	Packaging plus logistics is 9.1% of part cost. Freight is bought by the cubic metre — parts that nest, stack or ship in denser dunnage cut it directly.	¥0.43	2.7%	Quick Win	Low

§13 — Opportunity by Category

4 categories with an actionable lever

Category	Opportunities	Best single action	Best save/part	Category total (indicative)
Tooling	3	Volume Increase to Dilute Tooling NRE	¥1.59	¥2.86
Design & Geometry	1	Design Review for DFM Compliance	¥0.79	¥0.79
Commercial & Sourcing	1	Annual Volume Re-Commitment for Better Pricing	¥0.48	¥0.48
Packaging & Logistics	1	Pack Density / Cube Utilisation Review	¥0.43	¥0.43

§15 — Implementation Roadmap

Quick Wins — Implement Immediately

- Volume Increase to Dilute Tooling NRE
- Tooling Ownership and End-of-Life Terms
- Pack Density / Cube Utilisation Review
- Design Review for DFM Compliance
- Annual Volume Re-Commitment for Better Pricing